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Maternal Attention Deficit Hyperactivity Disorder and Perinatal Depression: A Systematic Review

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Natalie Pierotti[1] and Kaushadh Jayakody[1,2,3]

Abstract

Background: Attention deficit hyperactivity disorder (ADHD) is well-established as a disorder that persists into adulthood and is commonly comorbid with other mental health conditions. Despite increasing numbers of women entering reproductive years with this diagnosis, there is limited research on it in the perinatal period.

Objective: To critically appraise the available literature and examine the association between maternal ADHD and depression in the perinatal period.

Methods: A systematic review was conducted by searching EMBASE, PsycINFO, MEDLINE and Google Scholar up to August 2024. The available literature was evaluated in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PRISMA). The quality assessment was performed using the Joanna Briggs Institute framework, and a narrative synthesis was completed.

Results: Ten studies were identified as meeting the inclusion criteria. The majority were of medium quality, with issues identified in sample size, study design and validity of the self-report scale used. Our review found a prevalence of 16.76% to 57.6% of perinatal depression in women with ADHD. An increased risk of developing perinatal depression was also identified in women with ADHD (OR 1.8–2.63) compared to those without the disorder. Finally, our review also found a positive association between ADHD symptoms and depressive symptoms in perinatal women.

Conclusion: Our findings suggest a much greater vulnerability to perinatal depression among women with ADHD, and an association between ADHD symptoms and depressive symptoms in the perinatal period. There is, however, a clear need for more high-quality studies to replicate and further explore these findings. (*J. of Att. Dis. 2026; 30(3) 300-314*)

Keywords
ADHD, depression, perinatal, antenatal, postnatal

Introduction

Attention deficit hyperactivity disorder (ADHD) is a common neurodevelopmental disorder with an estimated prevalence of 5.9% to 7.1% in youth populations (Willcutt, 2012) and 2.5% of adults (Simon et al., 2009). Core symptoms include inattention, hyperactivity and impulsivity (American Psychiatric Association, 2013). Once considered a childhood condition, there is now increasing recognition that symptoms persist into adulthood with significant costs at both an individual and societal level. Adults with ADHD have been shown to under-achieve in terms of education and employment compared to their intellectual abilities (Kaiser et al., 2008). They also report more impairment in interpersonal relationships relative to adults without ADHD (Safren et al., 2010). The economic burden of ADHD is estimated to be between $US356 million to 20.27 billion for national estimates (Chhibber et al., 2021).

Attention deficit hyperactivity disorder is commonly comorbid with other mental disorders, including depression, anxiety and substance use disorders (Choi et al., 2022). Depression, in particular, has a strong association with ADHD both in childhood and as an adult (Chang et al., 2016; Meinzer et al., 2013, 2014). It has been estimated that 44% of individuals with ADHD experience a depressive episode before age 30 compared to 25% of

[1]Bendigo Health, Bendigo, VIC, Australia
[2]Faculty of Medicine Nursing and Health Sciences, School of Rural Health, Monash University, Burwood, VIC, Australia
[3]Institute for Mental and Physical Health and Clinical Translation (IMPACT), School of Medicine, Deakin University, VIC, Australia

Corresponding Author:
Natalie Pierotti, Bendigo Health, Level 2, 100, Barnard Street, VIC 3551, Australia.
Email: npierotti@bendigohealth.org.au

individuals without ADHD (Meinzer et al., 2013). Importantly, women with ADHD tend to have more prolonged episodes of depression and are at greater risk of suicide (Biederman et al., 2008).

Twin studies suggest that the association between ADHD and depression co-occurrence is explained principally by shared genetic factors (Faraone & Larsson, 2019). However, it has also been proposed that ADHD causes depression through exposure to increased psychosocial stressors such as discordant relationships and poor academic attainment, which in turn increases the risk of depression (Riglin et al., 2021).

Attention deficit hyperactivity disorder has historically been under-recognised in women due to a different symptom profile, with less so-called disruptive behaviours. Masking and internalisation of symptoms may delay receiving a diagnosis and appropriate treatment (Fraticelli et al., 2022). This, in turn, leads to lower levels of self-esteem and self-efficacy (Rucklidge & Kaplan, 1997). Despite increasing numbers of women entering reproductive years with a diagnosis of ADHD, there is a paucity of research on the subject, particularly in the perinatal period, with most studies to date focused on the impact of medication in pregnancy and breastfeeding (Anderson et al., 2020; Bang Madsen et al., 2023; Källén et al., 2013; Li et al., 2020; Nörby et al., 2017).

The perinatal period, encompassing both the antenatal and postnatal periods, is defined in Australian clinical practice by the Royal Australian and New Zealand College of Psychiatrists as being from conception to 12 months postpartum (RANZCP, 2021). Other definitions vary (WHO, 1977); however, all agree it is a uniquely vulnerable time for the mother, infant, and the wider family. It is a time of increased stress, both physiological and psychosocially, and is likely to represent a high-risk time for women with ADHD, as with other chronic mental health conditions (Wesseloo et al., 2016). ADHD is known to be associated with an increased risk of unplanned pregnancy (Owens & Hinshaw, 2020) and pregnancy at an earlier age (Lehti et al., 2012), as well as obstetric complications such as preterm birth, preeclampsia and hyperemesis (Walsh et al., 2022).

Perinatal depression, encompassing both antenatal and postpartum phases, is well recognised. The prevalence of antenatal depression is estimated to be between 7% and 20% in high-income countries, and for postpartum depression between 7% and 30% across low, middle and high-income countries (Biaggi et al., 2016). Depression in the perinatal period is associated with significantly increased morbidity and mortality, including adverse birth outcomes such as preterm birth, low birth weight and stillbirth (Simonovich et al., 2021). Perinatal depression is also associated with disrupted attachment and adversely impacts the infant's emotional, social, motor, cognitive and behavioural development (Slomian et al., 2019). Significantly, suicide remains one of the most frequent causes of death in women in the postnatal period (Australian Institute of Health and Welfare, 2023).

Recent research has highlighted the role of hormones in ADHD (Dorani et al., 2021; Eng et al., 2024; Martel et al., 2009). It appears that rather than being perinatal in origin, females may have multiple critical developmental periods corresponding with reproductive life events. Symptom severity may also fluctuate across the menstrual cycle, corresponding to changes in oestrogen (Eng et al., 2024). It has been hypothesised that reproductive hormones also play a role in the pathophysiology of perinatal depression through the regulation of the hypothalamic-pituitary-adrenal axis and the production of neurotransmitters (Meltzer-Brody, 2011). It is, therefore, possible that the hormonal changes that occur in the postpartum period may play a causal role in the development of both perinatal depression and ADHD.

To the best of our knowledge, there are no systematic reviews examining the association between ADHD and depression in the perinatal period. Kittel-Schneider et al.'s (2021) systematic review examined the association between ADHD and adverse pregnancy and birth outcomes but did not address depression. Antoniou et al.'s (2022) literature review examines the association between maternal ADHD and parental distress more widely. Despite growing recognition of adult ADHD, its interplay with perinatal depression remains an underexplored area. Our review intends to address this deficit.

Our aim is to examine the association between maternal ADHD and depression in the perinatal period. Our objectives were to identify:

1. The rates or prevalence of perinatal depression in women with ADHD
2. If women with ADHD are at risk for developing perinatal depression
3. If there is a relationship/correlation between ADHD symptoms and depressive symptoms

Methodology

Search Strategy and Selection Criteria

To achieve these objectives, a systematic review was conducted after applying inclusion and exclusion criteria. Available literature was reviewed per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses criteria (PRISMA) (Page et al., 2021) across four databases: EMBASE, PsycINFO, MEDLINE and Google Scholar. EMBASE, PsycINFO and MEDLINE were searched from inception to August 2024, using a search strategy. For a detailed search strategy, see Supplementary Material S1. Google Scholar was searched using a modified search string 'Maternal ADHD and perinatal depression' and the first 20 pages of results were included. Studies were limited to peer-reviewed published articles written in English. Results were supplemented by searches of reference lists, citations and author searches.

Inclusion Criteria

- Any study based on original empirical data published in a peer-reviewed journal, in the English language, all years, with full-text availability.
- Where the timeframe of the study was within the perinatal period, defined as from conception to the end of the first year of life in keeping with Australian clinical standards (RANZCP, 2021).
- Where the study assessed maternal ADHD (including attention deficit disorder). This included studies in which participants were identified as having ADHD symptoms via a self-report scale, who were diagnosed with ADHD following a formal assessment, or who had a relevant Diagnostic Statistical Manual (DSM) or International Classification of Diseases (ICD) code assigned to them in their records.
- Where the study assessed depressive symptoms. This included studies in which participants were diagnosed with clinical depression, such as major depressive disorder or dysthymia, following DSM or Structured Clinical Interview for DSM Disorders (SCID) assessments, or who had a relevant DSM or ICD code assigned to them in their records. Studies that assessed depressive symptoms using self-report tools were also included.

Exclusion Criteria

- Qualitative studies, case reports, and systematic reviews
- Studies looking at the relationship between ADHD and depression outside of the perinatal period
- Studies examining the relationship between ADHD and other diagnoses during perinatal period

Outcome Measures

The outcome measures used were prevalence or prevalence ratio (PR) of perinatal depression in women with ADHD.

The main measures used for the risk of developing perinatal depression were odds ratio (OR) and or relative ratio (RR), depending on the availability of data.

Measures of association between ADHD and depressive symptoms were assessed using correlation coefficient and regression models.

Data Extraction and Qualitative Assessment

Search results were imported into Covidence software (Veritas Health Innovation, n.d.). All screening stages were completed on this platform. The lead author performed title and abstract screening to exclude studies that did not meet the inclusion criteria, engaging in discussions with the other author to address any uncertainties. For studies that did, the full article was sourced and assessed for relevance. The second author conducted a cross-check by independently reviewing 50% of the quality-screened papers to ensure they met criteria. Data was extracted from the articles and combined into tables where appropriate.

Quality appraisal of the included papers was conducted using the Joanna Briggs Institute (JBI) critical appraisal tool (Moola et al., 2017) for the relevant study type. For further details, see Supplementary Material S2. Papers were assigned a rating of good, medium or poor quality based on a consensus decision between the two authors. The outcome was reflective of the information included in the critical appraisal tool.

Given the limited number of papers found and the considerable variability in study design, statistical methods, quality and reporting, a full meta-analysis was not possible. Therefore, the results are presented as a systematic narrative review.

Results

Search Results

An initial search of the electronic databases returned 2,634 references. A total of 844 were excluded as duplicates by the Covidence software. 49 were assessed for eligibility at full text, and ten studies were included in the final review (see Figure 1).

Included in the final review (see Tables 1–3) are three cohort studies: one prospective (Baker et al., 2022), two retrospective (Andersson et al., 2023; Walsh et al., 2022) and seven cross-sectional studies (Dorani et al., 2021; Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009).

Of these 10 included studies, five (Baker et al., 2022; Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Ninowski et al., 2007; Walsh et al., 2022) examined the antenatal period, and four (Andersson et al., 2023; Dorani et al., 2021; Semple et al., 2011; Watkins & Mash, 2009) examined the postpartum period. One study (Walsh et al., 2022) appears to have examined both; however, the timeframes are ambiguous. Most of the studies came from the USA or Canada, with two (Andersson et al., 2023; Dorani et al., 2021) from Europe and one (Murray et al., 2022) looking at eight low and middle-income countries.

Two studies (Andersson et al., 2023; Dorani et al., 2021) reported the prevalence of depression in those with ADHD, two studies (Baker et al., 2022; Walsh et al., 2022) looked at the risk of developing depressive symptoms in women with ADHD and six studies (Jones, Eddy, Bourchtein, et al.,

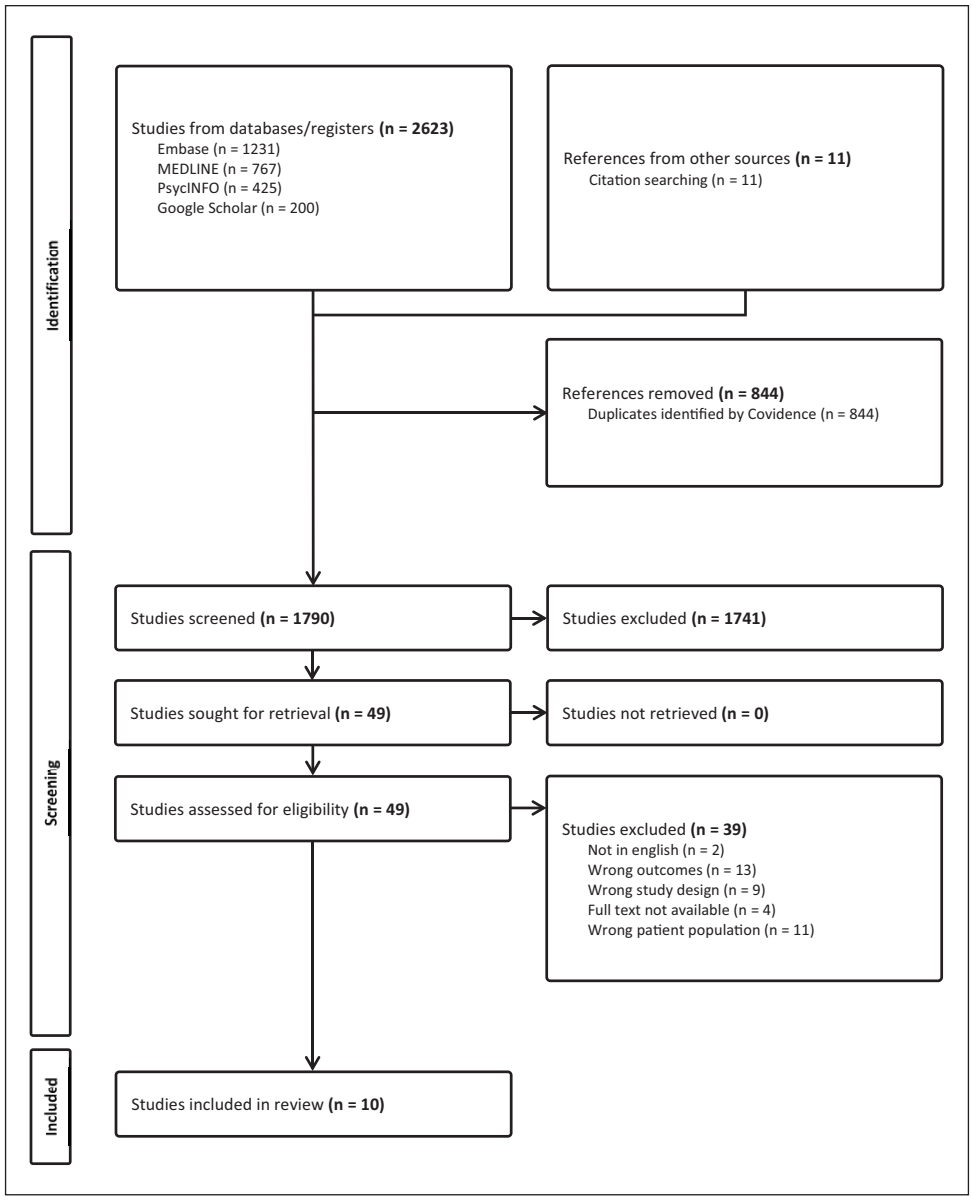


Figure 1. PRISMA search results summary flow chart.

2018; Jones, Eddy, Rabinovitch, et al., 2018; Murray et al., 2022; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) looked at the association between ADHD and depressive symptoms.

Methodological Quality of Reviewed Studies

Half of the included studies were rated medium based on JBI checklist criteria (see Supplementary Material S2), with only one study (Andersson et al., 2023) obtaining a good rating.

Two studies (Baker et al., 2022; Semple et al., 2011) were downgraded to poor as a result of having very low numbers included.

Significantly, nine studies (Baker et al., 2022; Dorani et al., 2021; Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Murray et al., 2022; Ninowski et al., 2007; Semple et al., 2011; Walsh et al., 2022; Watkins & Mash, 2009) did not adequately control for confounding variables. Six studies (Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Murray et al., 2022; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) were scored lower based on the validity or reliability of the self-report scale used; for example, the use of unvalidated scales, scales with limited data to support its use, or with low sensitivity in the perinatal population.

Table 1. Prevalence of Perinatal Depression in Women With ADHD.

Study	Design	Control group	Sample size (*N*) Country Setting	Antenatal/postnatal Weeks pregnant/ postpartum	Points of assessment during study	ADHD rating scale	Depression rating scales	Results	Quality
Andersson et al (2023)	Retrospective cohort study	Yes	*N* = 773,047 Sweden National Patient Registry Data	Postnatal N/A	N/A	ICD 9 or 10–ADHD Or prescribed ADHD medications prior to birth of child	ICD 8,9 or 10–Depression	The study reported a prevalence of depression of 16.76% in women with ADHD vs. 3.29% in women without. Prevalence ratio (PR) 5.09 (95% confidence interval (CI) [4.68, 5.54]).	Good
Dorani et al. (2021)	Cross-sectional study	No	*N* = 85 Netherlands Outpatient ADHD clinic	Postnatal Retrospective	Single point of assessment	Diagnostic interview DSM5 for ADHD	Retrospective EPNDS	57.6% met criteria for Postpartum depression in the postpartum period after the first childbirth.	Medium

Note. EPNDS = Edinburgh Postnatal Depression Scale; N/A = Not applicable.

Table 2. Risk of Developing Perinatal Depression in Women with ADHD.

Study	Design	Control group	Sample size (*N*) Country Setting	Antenatal/postnatal Weeks pregnant/ postpartum	Points of assessment during study	ADHD rating scale	Depression rating scales	Results	Quality
Walsh et al. (2022)	Retrospective cohort study	Yes	*N* = 85,832 USA Registry data TriNetX database	Both N/A	N/A	ICD 10 code – ADHD	ICD 10 code - Depression - Postpartum depression	2.1% of patients with an ADHD diagnosis also had a postpartum depression diagnosis vs. 1.2% in those without. OR 1.8 15.7% of patients with an ADHD diagnosis also had a depression diagnosis vs. 6.6% in those without. OR 2.63	Medium
Baker et al. (2022)	Prospective cohort study	No	*N* = 28 USA Community	Antenatal Planned pregnancy or <20/40	1) <20/40 2) 24/40 3) 36/40	Diagnostic interview DSM5 for ADHD AISRS	MADRS EPNDS	Women who discontinued psychostimulant treatment during pregnancy had a clinically significant increase in EPNDS scores. Adjusted mean EPNDS changes were: discontinuers, 4.32, meds as needed, −1.01, and maintainers, −0.65.	Poor

Note. AISRS = Adult ADHD Investigator Symptom Rating Scale; EPNDS = Edinburgh Postnatal Depression Scale; MADRS = Montgomery–Åsberg Depression Rating Scale.

Table 3. Association Between Maternal ADHD and Depressive Symptoms.

Study	Design	Control group	Sample size (N) Country Setting	Antenatal/ Postnatal Weeks pregnant/ postpartum	Points of assessment during study	ADHD rating scale	Depression rating scales	Results	Quality
Murray et al. (2022)	Cross-sectional data within a cohort study	No	$N = 1{,}204$ Eight- low- or middle-income countries Antenatal clinics	Antenatal 29–40/40	1) 3rd trimester 2) between 3 and 39 weeks postpartum	Own ADHD rating scale	PHQ-9	ADHD symptoms significantly associated higher levels of depressive symptoms ($b = 0.30$, $p < .001$) and stress ($b = .34$, $p < .001$) during pregnancy, even after correcting for multiple comparisons.	Medium
Ninowski et al. (2007)	Cross-sectional study	No	$N = 86$ Canada Antenatal clinics	Antenatal 2nd or 3rd trimester	Single point of assessment	CAARS ABCA	BSI-18	Depression remained significantly positively correlated with the CAARS ADHD Index, $r(81) = 0.23$, $p < .05$, as did anxiety, $r(81) = .32$, $p < .05$.	Medium
Semple et al. (2011)	Cross-sectional study	No	$N = 40$ Canada Community health centres	Postnatal Infants aged 3–8/12	Single point of assessment	ADHD quick screen ABCA	BSI-18	Maternal ADHD symptoms were positively correlated with T scores on the BSI-18 anxiety subscale ($r = .40$, $p < .05$) but not with scores on the depression and somatisation subscales or the GSI.	Poor
Jones, Eddy, Rabinovitch, et al. (2018)	Cross-sectional study	No	$N = 198$ USA Antenatal clinic	Antenatal Pregnant	Single point of assessment	CAARS	PHQ-2	Depressive symptoms correlated with all three ADHD symptom Clusters. Inattention ($r = .254$), hyperactivity ($r = .434$) and impulsivity ($r = .443$) $p < .01$	Medium/ poor
Watkins and Mash (2009)	Cross-sectional study	No	$N = 99$ Canada Immunisation clinics at community health centres	Postnatal Infants aged 6/12	Single point of assessment	CAARS	BSI-18	Co-occurring symptoms of depression were significantly related to ADHD symptom clusters. Global Severity Index ($r = .57$), inattention ($r = .42$), hyperactivity ($r = .23$) and impulsivity ($r = .56$) $p < .05$	Medium
Jones, Eddy, Bourchtein, et al. (2018)	Cross-sectional study	No	$N = 116$ Identifying as Black or Hispanic USA Antenatal clinic	Antenatal 5–40/40	Single point of assessment	CAARS AAQoL	PHQ-2	ADHD and depression risk were strongly positively correlated ($r = .54$, $p < .001$). In addition, depression risk was negatively correlated with the three AAQoL subscales with increased depression risk relating to decreased QoL.	Medium/ poor

Note. *r*-correlation (correlation coefficient). ABCA = ADHD Behaviour Checklist for Adults; AAQoL = Adult ADHD Quality of Life; BSI-18 = Brief Symptom Inventory; CAARS = Conner's Adult ADHD Rating Scale – Self Report, Short form; PHQ-9/2 = Patient Health Questionnaire.

Five studies (Dorani et al., 2021; Murray et al., 2022; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) also lacked sufficient demographic details.

Measurements Used in ADHD

Two retrospective cohort studies (Andersson et al., 2023; Walsh et al., 2022) used the inclusion of an ICD code specific to ADHD to define ADHD diagnosis.

Two studies (Baker et al., 2022; Dorani et al., 2021) completed structured clinical interviews for DSM5 diagnostic criteria for ADHD.

Murray et al. (2022) used a non-validated self-report scale consisting of a 5-item questionnaire measuring current ADHD symptoms. This was reported as an attempt to create a more sensitive tool to pick up ADHD in women but was found to have low internal consistency (Cronbach's alpha 0.59).

Semple et al. (2011) used the ADHD Quick Screen to inform recruitment but relied on the ADHD Behaviour Checklist for Adults (ABCA) in analysis. This scale measures childhood and current symptoms.

The remaining four studies (Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Ninowski et al., 2007; Watkins & Mash, 2009) used the Conner's Adult ADHD Rating Scale – Self Report, Short form (CAARS). This scale measures current symptoms via self-report and informant report.

Measurements Used for Depression/Depressive Symptoms

The two retrospective cohort studies (Andersson et al., 2023; Walsh et al., 2022) used an ICD code as evidence of a diagnosis of clinical depression. All other included studies used self-report scales rather than clinical interviews.

Two studies (Baker et al., 2022; Dorani et al., 2021) used the Edinburgh Postnatal Depression Scale (EPNDS). This is the most widely used screening tool for postnatal depression (Park & Kim, 2023).

Three studies (Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) used the Brief Symptom Inventory (BSI-18). It includes three subscales: depression, anxiety and somatisation, as well as a Global Severity Index score (Hamidia et al., 2022). The depression subscales were used for the purpose of our review.

Murray et al. (2022) used the Patient Health Questionnaire-9 (PHQ-9). The PHQ-9 is the most widely used depression measure globally. Its nine questions correspond to the DSM criteria for depression (Wang et al., 2021).

Finally, two studies (Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018) used the Patient Health Questionnaire -2 (PHQ-2). The PHQ-2 is an abbreviated version of the PHQ-9, containing only two questions that assess the frequency of depressed mood and anhedonia. (Gigantesco et al., 2021).

Primary Outcomes

Prevalence of Perinatal Depression in Women with ADHD. Two studies (Andersson et al., 2023; Dorani et al., 2021) examined the prevalence of perinatal depression in women with ADHD. The combined sample size was 773,132. Both studies reported on the postnatal period.

Andersson et al. (2023) completed a retrospective cohort study ($N=773,047$) using Swedish registry data. The study reported a prevalence of depression of 16.76% in women with ADHD compared to 3.29% in women without ADHD. This equated to a prevalence ratio (PR) of 5.09. This can be compared to existing estimates of the prevalence of postnatal depression, of 7% -30% in the general population with variation in results noted depending on the country and measure used (Biaggi et al., 2016). The result, however, is subject to possible surveillance bias as women with ADHD are more likely to be seen in psychiatric clinics and, therefore, have a diagnosis of depression made. Interestingly, the study had overall statistically low levels of ADHD compared to the general population (approximately 0.5%), which may represent the fact that ADHD is underdiagnosed in adult females. A study by Polyzoi et al. (2018) reported the estimated prevalence of ADHD in adulthood in Sweden of 0.58% in 2006, increasing to 3.55% in 2011, with a corresponding increase in the proportion of females with the diagnosis from 37.5% to 44.8%.

Dorani et al. (2021) examined the prevalence of hormone-related mood disorders in women with ADHD. Results from this study ($N=85$) showed a prevalence of 57.6% of women with ADHD meeting the criteria for postpartum depression after the birth of their first child. This is substantially higher than the general population estimates of between 7% and 30% across low, middle and high-income countries (Biaggi et al., 2016). The overall study followed 209 women with ADHD, however, only 85 were mothers and were therefore included in their analysis. This may have led to selection bias, as the study was also assessing other hormone-related mood disorders. Depression was also assessed retrospectively, making the results likely susceptible to recall bias.

Risk of Developing Depression. Two studies (Baker et al., 2022; Walsh et al., 2022) with a combined sample size of 85,860 investigated the risk of developing perinatal depression in women with ADHD.

Baker et al. (2022) stratified the risk of developing depression in the antenatal period based on patterns of ADHD medication use. This was a prospective cohort study ($N=28$) where the cohorts were defined based on patterns of medication use: discontinuers, maintainers, and a group who changed their regime with the aim of reducing use. The

study showed significant differences between those who discontinued medication versus those changing medication ($p < .0001$) and discontinuers versus maintainers ($p = .0009$). This suggests that women who discontinued psychostimulant treatment during pregnancy had a clinically significant increase in EPNDS scores. Although not explicitly stated, the overall average EPNDS score was low across all three groups, and participants would not have met the threshold for suspected depression. This may be why the Montgomery–Åsberg Depression Rating Scale (MADRS) results were not statistically significant. This was also a low-powered study ($N = 28$) and included high levels of education and full-time work within the study population, which may not be representative of the wider population of women with ADHD. Lower socioeconomic status and low levels of education are independent risk factors for developing perinatal depression (Yang et al., 2022).

Walsh et al. (2022) examined the risk of developing obstetric complications in women with ADHD. This study was a retrospective cohort study utilising the TriNetX database, which sources information from insurance claims and electronic hospital records. They looked at two cohorts of 42,919 women with and without ADHD matched based on age, race and ethnicity. The study found that 2.1% of women with an ADHD diagnosis also had a postpartum depression diagnosis versus 1.2% in those without ADHD ($p = .001$ OR 1.8). Furthermore, 15.7% of women with an ADHD diagnosis also had a depression diagnosis, versus 6.6% in those without ADHD ($p = .001$ OR 2.63). This could be compared to a study by Kiewa et al. (2022) which reported the most significant individual comorbidity among women with perinatal depression was ADHD ($p = .009$ OR 2.3).

The study also completed sub-analyses based on the type of medication prescribed. Compared to the cohort without any medications, women on stimulant medications had statistically lower rates of depression OR 0.733 (CI [0.66, 0.813], $p = .001$) and postpartum depression OR 0.727 (CI [0.605, 0.874], $p = .001$). This is consistent with previous studies that show that being prescribed ADHD medication was associated with a reduced risk for subsequent and concurrent depression (Chang et al., 2016). Concerns have been raised about the safety of ADHD medications in pregnancy, however, recent systematic reviews have shown that the overall risk is low (Bang Madsen et al., 2023; Li et al., 2020), and opinions are starting to reflect this (Scoten et al., 2024). Women on non-stimulant medications did not have statistically significant results for either depression or postpartum depression. The study concedes that this may be because several of the medications, including bupropion and atomoxetine, are also used to treat depression.

Walsh et al. (2022) have reported postpartum depression separately from depression, presumably to allow cases of antenatal depression to be included; however, as it did not clarify if a woman could obtain both diagnoses, the results are unable to be combined. It also seems probable that, given the significantly higher rates of depression, at least some cases of depression occurring postnatally have been coded as depression.

Association Between ADHD and Depressive Symptoms. Six studies (Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Murray et al., 2022; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) investigated the association between ADHD symptoms and depressive symptoms assessed via self-report screening measures. They had a combined sample size of 1,743. Two studies (Semple et al., 2011; Watkins & Mash, 2009) were postnatal, four studies (Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Murray et al., 2022; Ninowski et al., 2007) were antenatal, and all reported on cross-sectional data.

Murray et al. (2022) reported on an exploratory multicentre study encompassing eight low and middle-income countries ($N = 1{,}204$). Results from all sites were pooled, so there is no site-specific data. They found that ADHD symptoms are associated with higher levels of depressive symptoms on the PHQ-9 ($p < .001$) during pregnancy, even after correcting for multiple comparisons.

Jones, Eddy, Bourchtein, et al. (2018) included 116 black pregnant women as part of a larger study of prenatal behaviours, maternal psychopathology, and quality of life (QoL). ADHD and depression risk were strongly positively correlated ($p < .001$). In addition, increased depression risk corresponded to decreased QoL; however, they were unable to draw conclusions as to the nature of this relationship.

Three studies (Jones, Eddy, Rabinovitch, et al., 2018; Ninowski et al., 2007; Watkins & Mash, 2009) reported a positive correlation between depression and all the CAARS subscales. Ninowski et al. (2007) also performed a partial correctional analysis and found that depression remained significantly positively correlated with the CAARS ADHD Index ($p < .05$) when controlling for the other subscales on the BSI-18.

The positive association found is consistent with existing research supporting an association between ADHD and depression both in children and adults (Biederman et al., 2008; Chang et al., 2016; Meinzer et al., 2014; Riglin et al., 2021).

The only negative study was Semple et al. (2011) which found maternal ADHD symptoms were positively correlated with the BSI-18 anxiety subscale ($p < .05$) but not with scores on the depression subscales ($N = 40$). However, Semple et al. (2011) included only a small sample size of 40 women with low overall levels of ADHD; only five mothers scored above the clinical cut-off.

Two of the studies (Jones, Eddy, Bourchtein, et al., 2018; Murray et al., 2022) included depression as a primary outcome, while the remaining four studies (Jones, Eddy,

Rabinovitch, et al., 2018; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) included it as a confounding factor to be controlled for while exploring other aspects of parenting, such as maternal-child attachment. While this still provides valuable evidence, it should be interpreted with caution, as relevant confounding factors for depression were not included.

Discussion

Summary of Findings

Our systematic review was undertaken to explore the links between maternal ADHD and perinatal depression. Ten studies were found and included in our review. Two studies looked at the prevalence of perinatal depression in women with ADHD, two studies examined whether women with ADHD are at risk of developing perinatal depression, and six studies looked at the association between ADHD and depressive symptoms in the perinatal period.

We found an increased prevalence (16.7%–57.6%) of perinatal depression in women with ADHD compared to the general population (Biaggi et al., 2016). An increased risk of developing perinatal depression was also identified in women with ADHD (OR 1.8–2.63) compared to those without. Our review also found a positive association between ADHD symptoms and depressive symptoms in perinatal women.

Discussion

In examining the relationship between ADHD and perinatal depression the studies included in this review are not able to make any causative statements. They do however suggest some potential contributing factors.

Baker et al. (2022) suggested that women with ADHD who discontinued psychostimulant treatment during pregnancy had significant impairments in family functioning. Specifically, that they are more likely to experience conflict within the family, have a harder time having fun as a family, rate parenting as more difficult, and describe being more isolated from their family. Murray et al. (2022) also found that ADHD symptoms in the third trimester of pregnancy were associated with lower social support from family, friends and significant others.

Ninowski et al. (2007) found that ADHD symptoms were significantly negatively correlated with prenatal expectations about what the baby will be like, enjoyment of motherhood, life as a mother, and abilities as a mother. ADHD symptoms significantly predicted negative reports of maternal self-efficacy. Watkins and Mash (2009) also found that ADHD symptoms significantly predicted parenting satisfaction after controlling for co-occurring symptoms of psychological distress. Jones, Eddy, Bourchtein, et al. (2018) found moderate to large negative correlations between maternal ADHD symptoms, depression risk and quality of life; defined as life outlook, productivity and relationships.

Women high on ADHD symptoms were also less likely to report that they wanted to get pregnant at the time they became pregnant. ADHD symptoms were also negatively related to feelings of happiness when participants found out they were pregnant according to Ninowski et al. (2007). Murray et al. (2022) also found an increased likelihood of an unwanted pregnancy.

ADHD symptoms were significantly negatively correlated with attendance at recommended prenatal checkups according to Ninowski et al. (2007) while Jones, Eddy, Rabinovitch, et al. (2018) showed that ADHD was positively associated with many of the prenatal health behaviours including poor eating and increased physical strain. Finally, Walsh et al. (2022) demonstrated higher rates of negative pregnancy outcomes in women with ADHD such as hyperemesis, eclampsia, pre-eclampsia, preterm birth, gestational diabetes and early pregnancy haemorrhage.

This constellation of decreased support, lower levels of self-efficacy and parenting satisfaction, higher levels of stress and negative birth outcomes will likely contribute to an increased risk for depression in women with ADHD. The broader evidence base supports also this hypothesis, that depression arises in women with ADHD as a result of being exposed to higher levels of psychosocial stressors (Riglin et al., 2021).

Outside of the perinatal period women with ADHD have already been demonstrated to experience lower levels of educational achievement (Fredriksen et al., 2014), and higher levels of discordant relationships (Barkley et al., 2007; Faraone & Biederman, 1997) which will predispose them to developing depression. Additional risk factors within the perinatal period includes the fact that women with ADHD are more likely to have an unplanned or unwanted pregnancy (Owens & Hinshaw, 2020), and pregnancy at a younger age (Lehti et al., 2012), likely due to higher levels of disorganisation and impulsive behaviours.

Finally, it is important to recognise that the perinatal period is a time of increased stress, and women with ADHD may find the adjustment more challenging due to the increased requirements to be organised and manage time efficiently. These executive functioning difficulties, when combined with difficulties in emotional regulation, self-esteem and sensory processing will create challenges when becoming a parent.

The current literature also suggests a biological basis for the relationship between ADHD and depression. This is supported by the results of Andersson et al. (2023) which found higher levels of teen pregnancies/younger age of mother, decreased cohabitation levels, lower levels of education and higher levels of co-morbid psychiatric

conditions in keeping with previous results. However, even after adjusting for these factors, estimates showed an independent association between ADHD, depression, and anxiety disorders postpartum.

Early family studies suggest that the co-occurrence of ADHD and depression is influenced by shared familial factors (Cole et al., 2009). More recent twin studies suggest that ADHD and depression co-occurrence is primarily explained by shared genetic factors (Cole et al., 2009; Faraone & Larsson, 2019). While a Genome-Wide Association Study of ADHD reported significant genetic correlation with various psychiatric disorders, the strongest of which was with depression (Demontis et al., 2019).

Riglin et al. (2021) used two different designs, a longitudinal population cohort design and Mendelian randomisation to investigate a possible causal relationship between ADHD and depression. Their findings suggest that ADHD does increase the risk of depression later in life and are consistent with a causal effect of ADHD genetic liability on subsequent major depression.

In addition to this genetic association, several recent papers have examined the possible contributions of sex hormones in the development, and symptom expression of both ADHD (Dorani et al., 2021; Eng et al., 2024; Martel et al., 2009) and depression in women (Meltzer-Brody, 2011). Current theories suggest that hormones such as oestrogen and progesterone have both organisational and activational effects on neurotransmitters including dopamine, serotonin and noradrenaline. Dopamine is recognised as a key neurotransmitter implicated in ADHD pathology (Martel et al., 2009).

As well as modulating neurotransmitters, sex hormones play a prominent role in regulation of stress and the hypothalamic-pituitary-adrenal (HPA) axis including at the hypothalamic corticotropin releasing hormone gene and multiple interactions between the estrogen and glucocorticoid receptors. As such they are also implicated in the development of post-partum depression. During third trimester of pregnancy, levels of oestrogen and progesterone rise dramatically, accompanied by activation of the HPA axis and increased levels of cortisol. After parturition the levels rapidly drop and the HPA axis must quickly compensate (Patterson et al., 2024).

Eng et al. (2024) proposes a similar theory of activation effects of cyclical circulating ovarian hormones on ADHD with increasing risk at times of rapid declines in oestrogen. While the evidence remains limited at this time, a recent systematic review concluded that a relationship between sex hormones and ADHD symptoms in females is likely, specifically in puberty and across the menstrual cycle (Osianlis et al., 2025).

Quality and Sources of Bias

There was substantial variability in the quality of the included studies; although the majority were of medium quality, several areas were consistently identified as potential sources of bias, including study design, sample size, lack of control for confounding factors and use of self-report scales.

Seven studies (Dorani et al., 2021; Jones, Eddy, Bourchtein, et al., 2018; Jones, Eddy, Rabinovitch, et al., 2018; Murray et al., 2022; Ninowski et al., 2007; Semple et al., 2011; Watkins & Mash, 2009) included in our review were cross-sectional studies, many of which either included multiple outcomes or did not look at depression as a primary outcome. All, with the exception of Baker et al. (2022), used a single point of assessment and were therefore unable to investigate changes in symptoms across the perinatal period. This will significantly limit the ability to draw conclusions about the direction of causality in the relationship or to determine periods of higher risk. Longitudinal studies would be required to elicit this information.

Importantly, Andersson et al. (2023) was the only study to attempt to control for relevant confounding factors. While many of the studies reported on comorbid psychiatric conditions, they did not include it in the analysis. This limits the validity of the results, given that many of these factors are independent risk factors for the development of perinatal depression (Yang et al., 2022).

Five out of six of the association studies had low or subthreshold levels of ADHD symptoms within their study populations based on self-report. This will limit the ability to detect a true effect. The exception was Jones, Eddy, Bourchtein, et al. (2018), in which 19.9% of the study population reported clinically significant symptoms of ADHD. It was, however, unclear how they obtained this.

A variety of measures were used to identify symptoms, and the lack of consistency across studies posed a challenge for comparing findings. Specific concerns should also be considered with the use of self-report scales. While most of the scales used have demonstrated acceptable sensitivity and specificity, screening tools are not meant to be utilised in isolation but are designed to be an adjunct to the gold standard structured clinical interview (Levis et al., 2019; Mulraney et al., 2022). Even for those studies utilising structured clinical interviews, no additional information was provided on how these were undertaken.

All the self-report scales utilise a Likert scale, which asks patients to assess how ‘often’ they manifest a symptom (Taylor et al., 2011). This is challenging to standardise as it is by nature subjective. Individuals with ADHD may have difficulty not only with accurately recalling their childhood symptoms, which would be subject to recall bias, but also with reporting their current symptoms. This may be due to limited self-awareness of behaviours that have been present since childhood, particularly externally manifested symptoms (Taylor et al., 2011). There is also an observed discrepancy between self and informer reports of symptoms, in particular inattentive symptoms (Zucker et al., 2002), which

affects reliability. Finally, Dunlop et al. (2018) noted that ADHD screening tools might have lower sensitivity and specificity in patients with major depressive disorder. In depressed patients, a negative screen strongly suggests the absence of ADHD, but a positive screen 'requires careful evaluation to determine whether self-reported ADHD symptoms simply emerge from depression or whether comorbid ADHD is present' (Dunlop et al., 2018). Given these issues, good quality ADHD scales should have both informant and self-rated versions for assessment of both adult and childhood symptoms (Barkley et al., 2002). Only one study, Ninowski et al. (2007), used partner reports of symptoms in addition to a self-report scale, and none of the studies provided data on childhood symptoms or age at which the diagnosis was made.

In terms of depression, the BSI-18 has very limited evidence for its use in the perinatal population. The PHQ-2 has also been shown to have low sensitivity in the perinatal population at the cut-off >3 (Gigantesco et al., 2021). This may be due to the fact that it is unable to account for those with mild depression, particularly in a low-prevalence population. Consideration should also be given to the cut-off scores used to determine depression. In studies utilising the EPNDS, the cut-off point for depression was set as 10/30, which indicates 'possible depression'. Gibson et al. (2009) suggested that a cut-off of 12 or 13 was likely to be more sensitive for a diagnosis of depression. They also suggested a higher cut-off for the antenatal population. This highlights the challenge of measuring symptoms in a diverse population across the perinatal period.

Strengths and Limitations

To the author's knowledge, this is the first systematic review to explore the association between maternal ADHD and the development of perinatal depression and, as such, fills an important gap in the clinical literature.

Our study has included both antenatal and postnatal populations; however, a limitation identified was the low number of studies able to be included. It is possible that additional papers may have been identified through grey literature searching or by directly contacting authors. Our review has, therefore, been unable to make any distinction between the various time periods or identify periods of particularly high risk.

Our review was also unable to combine results in a meta-analysis due to low numbers and lack of standardisation in population, diagnosis and measurements. Despite these issues, our review highlights the association between ADHD and depression, which has previously been well-established in a non-perinatal population.

Additional limitations include the fact that research has been mainly conducted in high-income countries, in community samples with no data provided from mother-baby units or similar inpatient units.

Clinical Implications

Our review has important implications for mental health clinicians and psychiatrists as well as midwives, obstetricians, maternal child health nurses, and GPs, all of whom are involved in the care of women during their childbearing journey.

Depression is already routinely screened for in clinical practice. The Centre of Perinatal Excellence (COPE, 2023) recommends that the EPNDS is completed at least once antenatally and once postnatally, generally at 6 weeks postpartum.

In the future, women with ADHD would benefit from early and more frequent screening for depression during the perinatal period to pick up on symptoms of concern.

Additional psychoeducation and support for the woman and her family would also be beneficial to optimise their mental health in the perinatal period. 'This includes education about the signs and symptoms of deteriorating mental health and strategies for supporting mental wellness' (Scoten et al., 2024).

For women scoring positive for depression, a brief screen for ADHD could be incorporated into clinical practice. If scoring positive on this, referral to a perinatal or adult mental health service for further evaluation would be indicated to allow for appropriate assessment and management.

To facilitate these outcomes, education for health professionals on ADHD and its presentation in women, the links with depression, and appropriate pathways and treatments in the perinatal period would be indicated.

Future Directions

Our review has demonstrated that for women with ADHD, perinatal depression is an important consideration. Despite the limitations in terms of quantity and quality of data, our review should be used to inform future research on the topic. Further large studies using gold standard measurements such as a self-rating scale (encompassing self and informer ratings) followed by a clinical interview and controlling for relevant confounding factors are required. This would allow for a future meta-analysis on the topic. Longitudinal studies with assessments at various time points across the antenatal and postnatal period would also be beneficial to investigate trends and high-risk times.

Conclusion

Our systematic review found an increased prevalence (16.7%–57.6%) of perinatal depression in women with ADHD compared to the general population (Biaggi et al., 2016). An increased risk of developing perinatal depression was also identified in women with ADHD (OR 1.8–2.63) compared to those without. Our review also found a positive

association between ADHD symptoms and depressive symptoms in perinatal women. There is, however, a clear need for more high-quality studies to replicate and further explore these findings.

With the growing recognition of adult ADHD, this review has important implications for clinical practice, as well as policy and service development for this vulnerable population. Our review highlights the importance of regular screening for depression in women with ADHD, as well as ensuring they have appropriate support to navigate the perinatal period successfully.

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ORCID iDs

Natalie Pierotti https://orcid.org/0009-0001-1079-9800
Kaushadh Jayakody https://orcid.org/0000-0002-5896-8538

Ethical Considerations

We confirm that this research consists of a systematic review, utilising only existing, published and non-identifiable records. It does not involve direct interaction with human subjects or their identifiable data. On these grounds ethical approval has been deemed to be not required.

Author Contributions

Natalie Pierotti: Conceptualisation; Methodology; Validation; Formal analysis; Investigation; Project administration; Resources; Visualisation; Writing – original draft; Writing – review & editing
Kaushadh Jayakody: Conceptualisation; Methodology; Project administration; Supervision; Resources; Writing – original draft; Writing – review & editing

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author, NP.

Supplemental Material

Supplemental material for this article is available online.

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Author Biographies

Natalie Pierotti is a Consultant Psychiatrist at Bendigo Health working in Perinatal and Adult Mental Health.

Kaushadh Jayakody is an Associate Professor (Affiliate) at the Institute for Mental and Physical Health and Clinical Translation (IMPACT), School of Medicine, Deakin University, VIC, Australia, and a Consultant Psychiatrist at Bendigo Health, VIC, Australia.